Appendix

Intuition and Derivations

April 18, 2026

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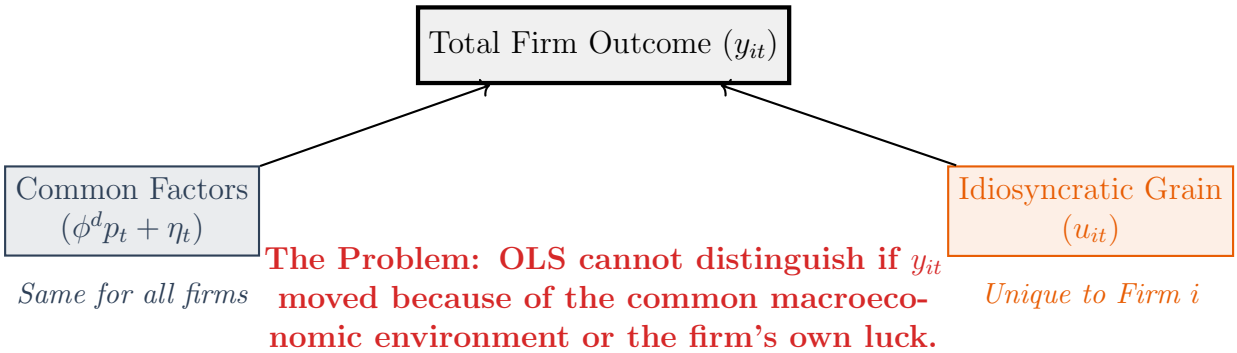
1 Visual Guide

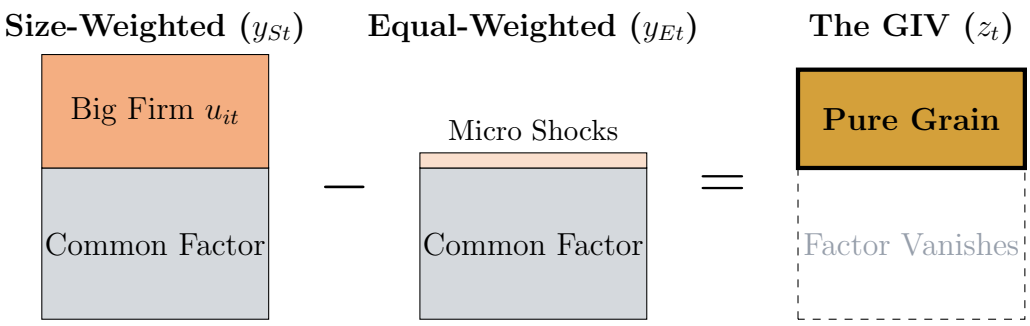
1.1 The Anatomy of a Firm Shock

For a visual learner, the demand equation $y_{it} = \phi^d p_t + \eta_t + u_{it}$ is a hierarchy of influences.



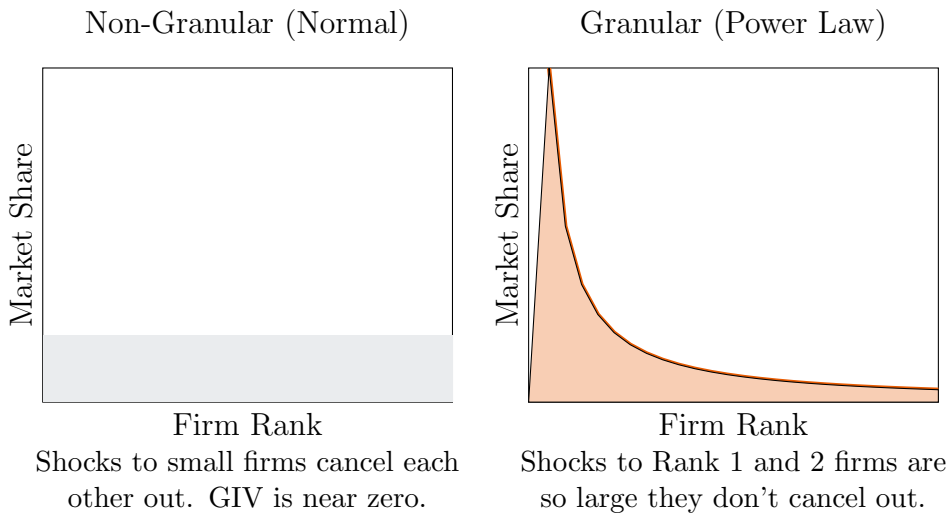
1.2 The "Differencing" Mechanism

The GIV ($z_t = y_{St} - y_{Et}$) acts as a filter. Imagine two "containers" of shocks. One is size-weighted (dominated by giants) and the other is equally-weighted (the crowd).



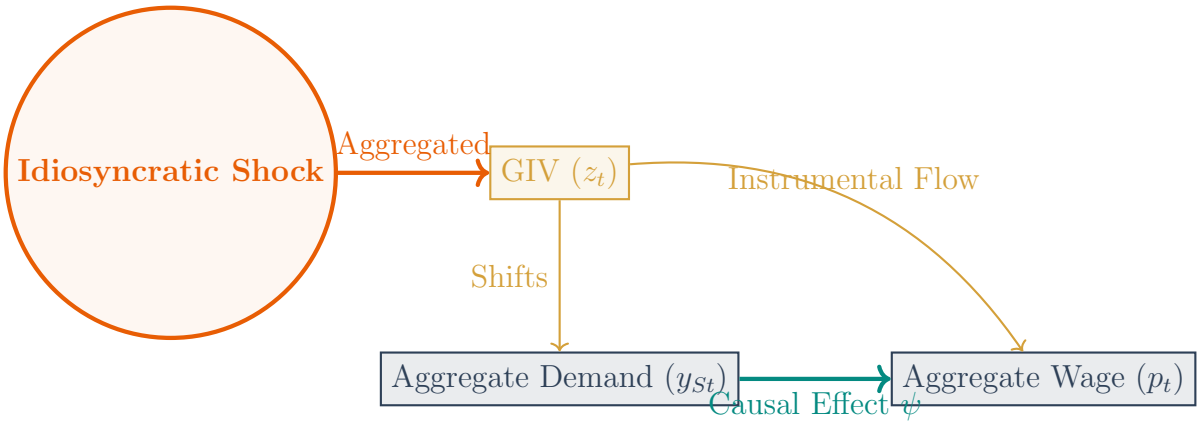
1.3 Why Granularity Matters (The Power Law)

GIV only works if the "Big Firm Shock" in the previous diagram is large enough to survive the averaging. In a "Normal" economy (left), the GIV is just noise. In a "Granular" economy (right), the giants move the needle.



1.4 Conceptual Flow of Identification

How does z_t help us find the causal effect ψ?



Identification Strategy: z_t is a "clean" shock that moves Aggregate Demand. By watching how Aggregate Wages respond to <u>only</u> that clean movement, we isolate ψ.

2 Derivation of equation 41

2.1 The Base Model

We start with the demand equation for firm i at time t, as given by Equation 39:

$$y_{it} = \phi^d p_t + \eta_t + u_{it} \tag{39}$$

where:

- y_{it} is the outcome for firm i.
- p_t is the common price (or wage).
- ϕ^d is the demand elasticity.
- η_t is a common aggregate shock.
- u_{it} is the idiosyncratic shock to firm i.

2.2 Deriving the Size-Weighted Aggregate (y_{St})

The size-weighted outcome is defined as $y_{St} = \sum_{i=1}^{N} S_i y_{it}$, where S_i are weights such that $\sum S_i = 1$.

Substituting Equation 39 into this definition:

$$y_{St} = \sum_{i=1}^{N} S_i(\phi^d p_t + \eta_t + u_{it})$$
$$y_{St} = \sum_{i=1}^{N} S_i \phi^d p_t + \sum_{i=1}^{N} S_i \eta_t + \sum_{i=1}^{N} S_i u_{it}$$

Since ϕ^d, p_t, and η_t do not depend on the index i, we can pull them out of the summations:

$$y_{St} = \phi^d p_t \left(\sum_{i=1}^{N} S_i \right) + \eta_t \left(\sum_{i=1}^{N} S_i \right) + \sum_{i=1}^{N} S_i u_{it}$$

Using the property $\sum S_i = 1$ and defining the size-weighted idiosyncratic shock as $u_{St} = \sum S_i u_{it}$, we get:

$$y_{St} = \phi^d p_t + \eta_t + u_{St} \tag{1}$$

2.3 Deriving the Equal-Weighted Aggregate (y_{Et})

The equal-weighted outcome is defined as $y_{Et} = \frac{1}{N} \sum_{i=1}^{N} y_{it}$.

Substituting Equation 39:

$$y_{Et} = \frac{1}{N} \sum_{i=1}^{N} (\phi^d p_t + \eta_t + u_{it})$$
$$y_{Et} = \frac{1}{N} \sum_{i=1}^{N} \phi^d p_t + \frac{1}{N} \sum_{i=1}^{N} \eta_t + \frac{1}{N} \sum_{i=1}^{N} u_{it}$$

Again, pulling constants out of the summations:

$$y_{Et} = \phi^d p_t \left(\frac{1}{N} \sum_{i=1}^{N} 1 \right) + \eta_t \left(\frac{1}{N} \sum_{i=1}^{N} 1 \right) + \frac{1}{N} \sum_{i=1}^{N} u_{it}$$

Since $\frac{1}{N} \sum_{i=1}^{N} 1 = \frac{1}{N} \cdot N = 1$, and defining the equal-weighted idiosyncratic shock as $u_{Et} = \frac{1}{N} \sum u_{it}$, we get:

$$y_{Et} = \phi^d p_t + \eta_t + u_{Et} \tag{2}$$

2.4 Result (Equation 41)

Combining these two results, we obtain Equation 41:

$$\begin{aligned} y_{St} &= \phi^d p_t + \eta_t + u_{St} \\ y_{Et} &= \phi^d p_t + \eta_t + u_{Et} \end{aligned} \tag{41}$$

3 Derivation of equation on slide 9

3.1 Introduction

Slide 9 of the presentation demonstrates how the Granular Instrumental Variable (z_t) can be used to identify both the supply elasticity (ψ) and the demand elasticity (ϕ^d). This document provides the underlying derivations and the statistical assumptions required for these results.

3.2 The Core Model and the GIV

Recall the system of equations from slide 5:

$$p_t = \psi y_{St} + \epsilon_t \tag{38}$$

$$y_{it} = \phi^d p_t + \eta_t + u_{it} \tag{39}$$

where $y_{St} = \sum S_i y_{it}$ is the size-weighted aggregate outcome. From slide 8, we define the Granular Instrumental Variable (z_t) as the difference between size-weighted and equal-weighted aggregate outcomes:

$$z_t = y_{St} - y_{Et} = u_{St} - u_{Et} \tag{42}$$

where $u_{St} = \sum S_i u_{it}$ and $u_{Et} = \frac{1}{N} \sum u_{it}$.

3.3 The Exogeneity Condition

The fundamental assumption of the GIV approach is that the idiosyncratic shocks u_{it} are independent of the aggregate shocks ϵ_t and η_t. Since z_t is a linear combination of only idiosyncratic shocks, it follows that:

$$\mathbb{E}[z_t \epsilon_t] = 0 \quad \text{and} \quad \mathbb{E}[z_t \eta_t] = 0 \tag{3}$$

This exogeneity allows z_t to serve as a valid instrument.

3.4 Identifying Supply Elasticity (ψ)

To find ψ, we use z_t as an instrument for y_{St} in the supply equation (38). We take the expectation of the product of the supply equation and the instrument:

$$\begin{aligned} \mathbb{E}[p_t z_t] &= \mathbb{E}[(\psi y_{St} + \epsilon_t) z_t] \\ \mathbb{E}[p_t z_t] &= \psi \mathbb{E}[y_{St} z_t] + \mathbb{E}[\epsilon_t z_t] \end{aligned}$$

By the exogeneity condition in (3), $\mathbb{E}[\epsilon_t z_t] = 0$. Thus:

$$\psi = \frac{\mathbb{E}[p_t z_t]}{\mathbb{E}[y_{St} z_t]} \tag{4}$$

This is the standard IV formula where z_t instruments for y_{St}.

3.5 Identifying Demand Elasticity (ϕ^d)

To find ϕ^d, we use the equal-weighted aggregate demand equation derived on slide 8:

$$y_{Et} = \phi^d p_t + \eta_t + u_{Et} \tag{41}$$

We aim to show that $\mathbb{E}[(y_{Et} - \phi^d p_t)z_t] = 0$. Substituting the demand equation:

$$\mathbb{E}[(y_{Et} - \phi^d p_t)z_t] = \mathbb{E}[(\eta_t + u_{Et})z_t] = \mathbb{E}[\eta_t z_t] + \mathbb{E}[u_{Et} z_t] \tag{5}$$

We already know $\mathbb{E}[\eta_t z_t] = 0$. Now we examine $\mathbb{E}[u_{Et} z_t]$. Under the assumption that u_{it} are i.i.d. with variance σ_u^2:

$$\begin{aligned}
\mathbb{E}[u_{Et} z_t] &= \mathbb{E}[u_{Et}(u_{St} - u_{Et})] \\
&= \mathbb{E}[u_{Et} u_{St}] - \mathbb{E}[u_{Et}^2] \\
&= \mathbb{E}\left[\left(\frac{1}{N}\sum_i u_{it}\right)\left(\sum_j S_j u_{jt}\right)\right] - \mathbb{E}\left[\left(\frac{1}{N}\sum_i u_{it}\right)^2\right] \\
&= \frac{1}{N}\sum_i S_i \sigma_u^2 - \frac{1}{N^2}\sum_i \sigma_u^2 \\
&= \frac{1}{N}\sigma_u^2(1) - \frac{1}{N}\sigma_u^2 = 0
\end{aligned}$$

Since $\mathbb{E}[u_{Et} z_t] = 0$, it follows that $\mathbb{E}[(y_{Et} - \phi^d p_t)z_t] = 0$. Rearranging:

$$\begin{aligned}
\mathbb{E}[y_{Et} z_t] &= \phi^d \mathbb{E}[p_t z_t] \\
\phi^d &= \frac{\mathbb{E}[y_{Et} z_t]}{\mathbb{E}[p_t z_t]}
\end{aligned}$$

This identifies the demand elasticity using the same granular instrument z_t.

4 Full model deep dive

4.1 Introduction

Slides 11 and 12 move from the "Simple Model" (where every firm responds identically to the macroeconomy) to the "Full Model." This is crucial because, in reality, superstar firms often have different sensitivities to interest rates, exchange rates, or global demand than small firms.

4.2 Slide 11: The Full Factor Model

4.3 The Equation

We start with the demand equation for firm i:

$$y_{it} = \phi p_t + \lambda_i \eta_t + u_{it} \tag{6}$$

where λ_i is the **factor loading**.

4.4 Intuition: What is a Factor Loading?

A factor loading λ_i measures firm i's *sensitivity* to a common macro shock η_t.

- **Example:** If η_t is a surge in energy prices, an airline like Delta might have a high λ_i (very sensitive), while a software firm like Microsoft might have a low λ_i.
- **Heterogeneity:** In the simple model, we assumed $\lambda_i = 1$ for everyone. In the full model, we acknowledge that large firms might be more (or less) sensitive to macro trends than the average firm.

4.5 Derivation: Why the Simple GIV Fails

The simple GIV is defined as $z_t^{simple} = y_{St} - y_{Et}$. Let's derive its components using Equation 6:

1. **Size-weighted average (y_{St}):**

$$\begin{aligned} y_{St} &= \sum S_i y_{it} = \sum S_i(\phi p_t + \lambda_i \eta_t + u_{it}) \\ &= \phi p_t \underbrace{\sum S_i}_{1} + \eta_t \underbrace{\sum S_i \lambda_i}_{\lambda_S} + \underbrace{\sum S_i u_{it}}_{u_{St}} \\ &= \phi p_t + \lambda_S \eta_t + u_{St} \end{aligned}$$

2. **Equal-weighted average (y_{Et}):**

$$\begin{aligned} y_{Et} &= \frac{1}{N} \sum y_{it} = \phi p_t + \eta_t \underbrace{\frac{1}{N} \sum \lambda_i}_{\lambda_E} + u_{Et} \\ &= \phi p_t + \lambda_E \eta_t + u_{Et} \end{aligned}$$

3. **The Resulting GIV:**

$$z_t^{simple} = y_{St} - y_{Et} = (\lambda_{\mathbf{S}} - \lambda_{\mathbf{E}})\eta_{\mathbf{t}} + (u_{St} - u_{Et}) \tag{7}$$

The Problem: If large firms have different macro-sensitivities than small firms ($\lambda_S \neq \lambda_E$), then the instrument z_t^{simple} is correlated with the common shock η_t. It is no longer a "pure" idiosyncratic shock; it is "leaky."

4.6 Slide 12: The Solution (Optimal Weights)

4.7 The Goal: Orthogonality

We want to choose a new set of weights, Γ_i, to construct a valid GIV: $z_t = \sum \Gamma_i y_{it}$. For this to be a valid instrument for the supply equation, it must be uncorrelated with the macro shock η_t.

4.8 Derivation of the Orthogonality Condition

Substitute Equation 6 into the definition of the weighted GIV:

$$\begin{aligned} z_t &= \sum \Gamma_i(\phi p_t + \lambda_i \eta_t + u_{it}) \\ &= \phi p_t(\sum \Gamma_i) + \eta_t(\sum \Gamma_i \lambda_i) + \sum \Gamma_i u_{it} \end{aligned}$$

To ensure z_t contains *only* idiosyncratic shocks (u_{it}), we must satisfy two conditions:

1. $\sum \Gamma_i = 0$: This removes the endogenous price p_t.
2. $\sum \Gamma_i \lambda_i = 0$: This removes the common macro shock η_t.

4.9 Why not just Firm Size (S_i)?

- **Power vs. Validity:** Firm size (S_i) gives the instrument its **power**. We want the instrument to be large, and only large firms can create aggregate "granular" ripples.
- **The Conflict:** If we use $z_t = \sum(S_i - 1/N)y_{it}$, we have power, but we lack validity if S_i is correlated with λ_i (e.g., if big firms are more sensitive to macro shocks).
- **The "Tilt":** We start with firm sizes S_i but then "tilt" or project them until they are perpendicular (orthogonal) to the macro-sensitivities λ_i.

4.10 Mathematical Solution (Projection)

In matrix form, we want Γ to be as close as possible to S but satisfy $\Gamma \perp \lambda$. This is a standard projection problem:

$$\Gamma = S - \lambda(\lambda'\lambda)^{-1}\lambda' S \tag{8}$$

Intuition: We take the firm sizes and subtract the portion of size that is explained by macro-sensitivity. What remains is "Pure Granular Weight."

4.11 Summary

- **Factor Loadings (λ_i)**: Firm-specific sensitivities to the macroeconomy.
- **Validity**: We cannot use raw size if size correlates with macro-sensitivity.
- **Optimal Weights (Γ_i)**: The "tilted" versions of size weights that satisfy $\sum \Gamma_i \lambda_i = 0$.

5 Gamma intuition

5.1 Recap: The Simple vs. The Full Model

In the simple model, we assume all firms respond identically to macro shocks. The GIV was simply:

$$z_t^{simple} = \sum(S_i - 1/N)y_{it} \tag{9}$$

This works because $S_i - 1/N$ is orthogonal to a constant factor (where every firm has $\lambda_i = 1$). However, when firms have heterogeneous sensitivities λ_i, the simple weights fail. We need Γ.

5.2 Intuition: Γ as "De-biased" Firm Size

5.3 The Simple GIV as a Special Case

Think of the simple GIV weights as $\Gamma_i^{simple} = S_i - \bar{S}$. We take the raw size and subtract the mean size. This "centers" the weights around zero, which kills any factor that hits everyone equally.

5.4 The Problem with Raw Size

If large firms are systematically more sensitive to interest rates (a common factor η_t), then the size-weighted outcome y_{St} will move whenever interest rates move. If we use y_{St} as an instrument, we are accidentally instrumenting with interest rate shocks, not idiosyncratic firm shocks.

5.5 The Logic of Γ

Γ is the part of firm size that is **unrelated** to macro-sensitivity.

- We ask: "How much of this firm's size is explained by its factor loading λ_i?"
- We subtract that "explained" part from the actual size.
- The result, Γ_i, is the "Pure Granular Weight."

If λ_i is completely uncorrelated with S_i, then Γ will look almost exactly like the simple GIV weights. If they are correlated, Γ "tilts" the weights to protect the instrument's validity.

5.6 The Math: Γ as a Projection Residual

Mathematically, we want to find weights Γ that are as close as possible to the size weights S (for power), but are orthogonal to the loadings λ (for validity).

We solve the following optimization problem:

$$\min_{\Gamma} \sum_{i=1}^{N} (\Gamma_i - S_i)^2 \quad \text{subject to} \quad \sum \Gamma_i = 0, \quad \sum \Gamma_i \lambda_i = 0 \tag{10}$$

The solution to this problem is equivalent to the **residual** of a linear regression. If we regress the vector of sizes S on a constant and the vector of loadings λ:

$$S_i = \alpha + \beta \lambda_i + \text{residual}_i \tag{11}$$

Then the optimal weight Γ_i is exactly that residual.

5.7 General Matrix Formula

If there are multiple factors (a matrix Λ), the formula is:

$$\Gamma = M_{\tilde{\Lambda}} S = (I - \tilde{\Lambda}(\tilde{\Lambda}'\tilde{\Lambda})^{-1}\tilde{\Lambda}')S \tag{12}$$

where $\tilde{\Lambda} = [\mathbf{1}, \Lambda]$ is the matrix of loadings plus a constant. $M_{\tilde{\Lambda}}$ is the "annihilator matrix" that projects S into the space orthogonal to the factors.

5.8 Summary

The transition from z^{simple} to z^{Γ} is a move from **demeaning** to **orthogonalizing**. In the simple model, we demean by subtracting $1/N$. In the full model, we orthogonalize by subtracting the projection of size onto macro-loadings.

6 Derivation of Gamma

6.1 The Optimization Problem

The goal of finding optimal GIV weights Γ is to stay as close as possible to the power-maximizing size weights S, while ensuring orthogonality to common factors (captured by loadings Λ) and a constant (to ensure the weights sum to zero).

Mathematically, we solve:

$$\min_{\Gamma} \frac{1}{2} \sum_{i=1}^{N} (\Gamma_i - S_i)^2 \tag{13}$$

subject to:

$$\sum_{i=1}^{N} \Gamma_i = 0 \qquad \text{(Constraint 1: Sum to Zero)}$$

$$\sum_{i=1}^{N} \Gamma_i \lambda_i = 0 \qquad \text{(Constraint 2: Orthogonality to } \lambda\text{)}$$

6.2 Lagrangian Setup

We define the Lagrangian $\mathcal{L}$ with multipliers μ_1 and μ_2:

$$\mathcal{L}(\Gamma, \mu_1, \mu_2) = \frac{1}{2} \sum_{i=1}^{N} (\Gamma_i - S_i)^2 + \mu_1 \left(\sum_{i=1}^{N} \Gamma_i \right) + \mu_2 \left(\sum_{i=1}^{N} \Gamma_i \lambda_i \right) \tag{14}$$

6.3 First Order Conditions (FOCs)

Differentiating with respect to the decision variable Γ_i:

$$\frac{\partial \mathcal{L}}{\partial \Gamma_i} = (\Gamma_i - S_i) + \mu_1 + \mu_2 \lambda_i = 0 \tag{15}$$

Rearranging for Γ_i, we get the structure of the solution:

$$\Gamma_i = S_i - (\mu_1 + \mu_2 \lambda_i) \tag{16}$$

6.4 Equivalence to Linear Regression

Recall the standard OLS model where we regress S_i on a constant and λ_i:

$$S_i = \alpha + \beta \lambda_i + \epsilon_i \tag{17}$$

The OLS estimator $(\hat{\alpha}, \hat{\beta})$ minimizes the sum of squared residuals $\sum \epsilon_i^2$. The resulting residuals are defined as:

$$\hat{\epsilon}_i = S_i - (\hat{\alpha} + \hat{\beta}\lambda_i) \tag{18}$$

Comparing this to Equation 16, we see that Γ_i has the exact same functional form as the OLS residual $\hat{\epsilon}_i$, where the Lagrange multipliers μ_1 and μ_2 play the roles of the OLS coefficients $\hat{\alpha}$ and $\hat{\beta}$.

6.5 Why the Constraints Matter

The "Normal Equations" that solve OLS are:

1. $\sum \hat{\epsilon}_i = 0$ (The sum of residuals is zero if an intercept is included).
2. $\sum \hat{\epsilon}_i \lambda_i = 0$ (Residuals are orthogonal to the regressors).

These are exactly our two constraints! Therefore, the values of μ_1 and μ_2 that satisfy our constraints are precisely the OLS coefficients $\hat{\alpha}$ and $\hat{\beta}$.

6.6 Conclusion

Solving for the optimal weights Γ is equivalent to regressing the vector of sizes S on the factor loadings (and a constant) and taking the residuals.

$$\Gamma = \text{Residuals of } (S \sim 1 + \Lambda) \tag{19}$$

In matrix notation, this is the projection:

$$\Gamma = M_X S = (I - X(X'X)^{-1}X')S \tag{20}$$

where $X = [\mathbf{1}, \Lambda]$. This projection removes the "macro-contaminated" portion of the firm size, leaving only the idiosyncratic granular component.